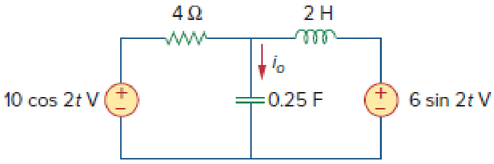
Problem

Solve for *io* in Fig. 10.73 using mesh analysis.

Figure 10.73:



Step-by-step solution

Step 1 of 6

$\omega = 2$

$10\cos(2t) \Rightarrow 10\angle 0°$

$6\sin(2t) \Rightarrow 6\angle -90°$

$6\angle -90° = -j6$

$2\,\text{H} = j\omega L$

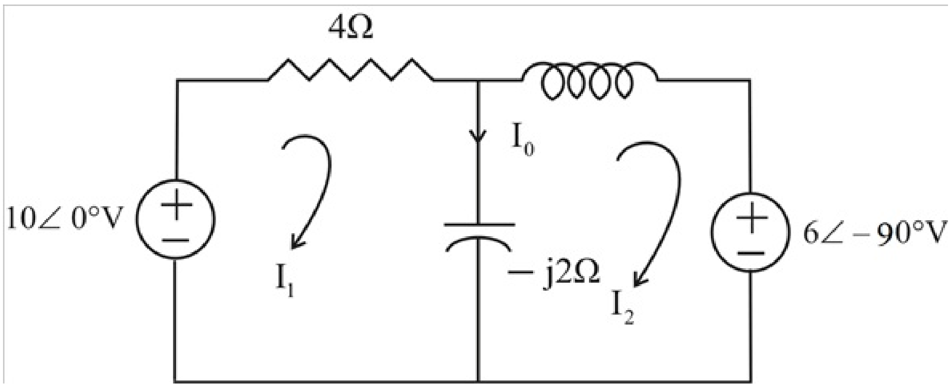
$j\omega L = 4j$

$0.25\,\text{F} \Rightarrow \frac{1}{j\omega C}$

$\frac{1}{j(2)\left(\frac{1}{4}\right)} = -j2$

Step 2 of 6

The circuit is shown below.



Step 3 of 6

For loop1,

$$-10+(4-j2)I_1+j2I_2=0$$

$$5=(2-j)I_1+jI_2 \text{ .... (1)}$$

For loop 2,

$$j2I_1+(j4-j2)I_2+(-j6)=0$$

$$j2I_1+(j2)I_2=j6$$

$$I_1+I_2=3 \text{ .... (2)}$$

Step 4 of 6

In matrix form (1) & (2) become,

$$\begin{bmatrix} 2-j & j \\ 1 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\Delta=2(1-j)$$

$$\Delta_1=5-j3$$

$$\Delta_2=1-j3$$

Step 5 of 6

$$I_1=\frac{\Delta_1}{\Delta}$$

$$I_1=\frac{5-j3}{2-j2}$$

$$I_1=2.062\angle 14°\text{ A}$$

$$I_2=\frac{\Delta_2}{\Delta}$$

$$I_2=\frac{1-j3}{2-j2}$$

$$I_2=1.118\angle -26.57°\text{ A}$$

Step 6 of 6

Therefore,

$$I_0=I_1-I_2$$

$$I_0=(2.0616\angle 14°)-(1.18\angle -26.57°)$$

$$I_o=1.414\angle 45°\text{ A}$$

$$\therefore \boxed{i_0(t)=1.414\cos(2t+45°)\text{ A}}$$